Sampling Effect on Performance Prediction of Configurable Systems: A Case Study (Artifact)

ABSTRACT

This document contains a stable URL to the artifacts associated with the paper "Sampling Effect on Performance Prediction of Configurable Systems: A Case Study" and provides general instructions to reproduce the presented results and reuse the datasets.

1 STABLE URL WITH THE ARTIFACTS

All resources, artifacts, and detailed procedures to replicate the paper results are available at github.com/jualvespereira/ICPE2020[1].

2 RESOURCES AND ARTIFACTS

The README.md file contains a detailed roadmap on the technical requirements of how to set up and run the experiments. It is structured into two parts: *(1)* performance prediction, and *(2)* aggregation and visualization of results.

2.1 Performance Prediction

To reproduce our results, users must follow the steps below:

- Install Docker: https://docs.docker.com/install/
- Download the container: `docker pull hmartinirisa/icpe2020:latest` (by invoking this script, all required ressources are installed, which might take several minutes).
- Run the container: `sudo docker run -it -v "$(pwd)":/docker hmartinirisa/icpe2020 bash`
- Go either to the directory `Distance-Based_Data_Time` or `Distance-Based_Data_Size`: `cd ICPE2020/Distance-Based_Data_Time` or `cd ICPE2020/Distance-Based_Data_Time`.
- For each `<sampling-approach>` (twise, solvBased, henard, distBased, divDistBased, rand) and `<case-study>` ($x264_0$, $x264_1$,..., $x264_{16}$), run the script: `./SPLConquerorExecuter.py <case-study> <sampling-approach> <save-location>` (example: `./SPLConquerorExecuter.py x264_0 rand /docker/ICPE2020/DistanceBased_Data_Time/SupplementaryWebsite/PerformancePredictions/AllExperiments`). The experiments will run for 100 random seeds.

Dependencies. The implementation depends on the SPL Conqueror for sampling and learning, and on the z3 Constraint solver to navigate through the configuration space of the configurable system.

Main repository. The main resources and artifacts of this paper are divided into two directories: `Distance-Based_Data_Time` and `Distance-Based_Data_Size`. Each directory contains the data related to *time* and *size* prediction, respectively. `SPLConquerorExecuter.py` is the main script to run the experiments and compute the prediction error rates. As input to this script, we need a feature model and the measurements of performance for the input video being considered. The directory `SupplementaryWebsite/MeasuredPerformanceValues` contains the data for all seventeen analysed input videos, namely $x264_0$, $x264_1$, $x264_2$, ..., $x264_{16}$.

- *FeatureModel.xml*: file containing the description of the configurable system $x264$.
- *measurements.xml*: file containing the measured performance values for all valid configurations of an specific input video.
- *input_video.txt*: file containing the description of the input video we used to measure all valid configurations.

To perform experiments related to a new case study *case_x*, the user must add the case-study name *case_x* into the files `SPLConquerorExecuter.py` and `analyzeRuns.py`, and add the files `FeatureModel.xml` and `measurements.xml` at a new folder *case_x* into the directory `Distance-Based_Data_Time/SupplementaryWebsite/MeasuredPerformanceValues` (as an example see these files for $x264$ case study – more information about the format of these files can be also found at github.com/se-passau/SPLConqueror).

The directory `SupplementaryWebsite/PerformancePredictions/AllExperiments` contains a set of prediction log files with the experiment results (*i.e.*, error rate). Each sampling error rate is computed 100 times using different random seeds for each input video.

- *learn_<sampling-approach>_<sampling-size>.a*: file containing a list of all input commands used to run SPL Conqueror[2].
- *out_<sampling-approach>_<sampling-size>.log*: file containing the error rate of applying a sampling strategy with a certain sample size on an input video (the error rate is the last number in the last line before *"Analyze finished"*).
- *sampledConfigurations_<sampling-approach>_<sampling-size>.csv*: file containing the set of configurations used as input sample to the machine-learning technique.

2.2 Aggregation and Visualization of Results

`analyzeRuns.py` and `ErrorRateTableCreator.py` are the main scripts to aggregate and visualize the results. `analyzeRuns.py` collects all error rates from all 100 runs of all case studies in an unique file *all_error_<sampling-approach>_<sampling-size>.txt* (see directory `PerformancePredictions/AllSummary`). Then, the script `ErrorRateTableCreator.py` reads this file and uses `PerformKruskalWallis.R` to generate the visualization data available in Tables 2-8 from the paper (see the output tex-files at the directory `latex`).

- `./analyzeRuns.py <run-directory> <output-directory>`
- `./ErrorRateTableCreator.py <input-directory> <sampling-approaches> <labels> <output-tex>`

`<run-directory>` is the directory where all runs of all case studies are stored. `<output-directory>` and `<input-directory>` are the directories where the aggregated results should be written to and read from, respectively[3]. `<sampling-approaches>` and `<labels>` contain the list of sampling approaches to consider and the labels that should be used in the table. `<output-tex>` contains the directory where the tex-files should be written to[4].

[1] These are proprietary artifacts (except the datasets with all computed measurements) from github.com/se-passau/Distance-Based_Data with slightly modifications.

[2] For detailed explanation on these commands see github.com/se-passau/SPLConqueror

[3] In the replication process, the generated error rates on this directory must be the same as the provided ones on the directory `AllSummary` of our GitHub repository for the same sampling approach, sample sizes and random seeds.

[4] see these command lines example at our repository (README.md).